Drew Lapeer

Astronomy Graduate Student
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Research interests: observational astrophysics, galaxy evolution, baryon cycle, physics of star formation & stellar feedback

EDUCATION

2024 – · · · ·	**PhD, MSc., University of Massachusetts–Amherst** Astronomy. Advisor: *Professor Daniela Calzetti*
2021 – 2024	**BSc., University of Michigan** Astronomy & Astrophysics (Honors), Interdisciplinary Physics. Thesis: *Dynamical SMBH Detection in Compact Galaxies with NIRSpec/IFU* Advisor: *Professor Monica Valluri*
2022 – 2023	**Washtenaw Community College** General Studies
2019 – 2021	**Macomb Community College** Michigan Transfer Agreement

RESEARCH POSITIONS

Start Feb. 2027	**Visiting Graduate Student Research Fellow,** IPAC/Caltech
2024 – · · · ·	**Graduate Research Assistant,** Department of Astronomy, University of Massachusetts–Amherst.
2022 – 2024	**Undergraduate Research Assistant,** Department of Astronomy, University of Michigan.

AWARDS

2026	**Code/Astro Travel Award**, Code/Astro Workshop ($1200)
2025	**Graduate Travel Support Award**, College of Natural Sciences, University of Massachusetts ($775)
2024	**Walter W. Wada Award for Community Engagement**, Department of Physics, University of Michigan ($800)

PUBLICATIONS

Publication list: NASA ADS
N (in-prep./under review): first-author: 1 (1); co-author as major contributor: 1 (1); co-author: 10 (1)
Total citations: 69 (2 first author); *h*-index: 5 (1 first author)

First/Second Author

1. **Drew Lapeer** et al. "FEAST: Probing Hierarchical Star Formation with the Spatial Distributions of Young Star Clusters". In: *The Astrophysical Journal* 999.2, 189 (Mar. 2026), p. 189. DOI: `10.3847/1538-4357/ae3a92`. arXiv: `2601.11434 [astro-ph.GA]`.
2. Behzad Tahmasebzadeh et al. "The Lower Limit of Dynamical Black Hole Masses Detectable in Virgo Compact Stellar Systems Using the JWST/NIRSpec IFU". In: *The Astrophysical Journal* 974.1, 60 (Oct. 2024), p. 60. DOI: `10.3847/1538-4357/ad6a1b`. arXiv: `2408.02142 [astro-ph.GA]`.

Co-Author

1. Anne S. M. Buckner et al. "The spatial evolution of star clusters in NGC 628 with JWST". In: *Monthly Notices of the Royal Astronomical Society* 545.3, staf2025 (Jan. 2026), staf2025. DOI: `10.1093/mnras/staf2025`. arXiv: `2511.11115 [astro-ph.GA]`.
2. Helena Faustino Vieira et al. "FEAST: A NIRSpec/MOS Survey of Emerging Young Star Clusters in NGC 628". In: *The Astrophysical Journal* 1001.2, 128 (Apr. 2026), p. 128. DOI: `10.3847/1538-4357/ae523f`. arXiv: `2603.09866 [astro-ph.GA]`.
3. Sean T. Linden et al. "Feedback in Extragalactic Star Clusters (FEAST): Spectral Energy Distributions and the Physical Properties of Star Clusters in NGC 628 with CIGALE". In: *in press at ApJ*, arXiv:2606.13391 (June 2026), arXiv:2606.13391. DOI: `10.48550/arXiv.2606.13391`. arXiv: `2606.13391 [astro-ph.GA]`.
4. Alex Pedrini et al. "The emerging timescale of young star clusters regulated by cluster stellar mass". In: *Nature Astronomy* (May 2026). DOI: `10.1038/s41550-026-02857-y`. arXiv: `2603.09867 [astro-ph.GA]`.
5. Giacomo Bortolini et al. "FEAST: JWST/NIRCam View of the Resolved Stellar Populations of the Interacting Dwarf Galaxies NGC 4485 and NGC 4490". In: *The Astrophysical Journal* 991.2, 212 (Oct. 2025), p. 212. DOI: `10.3847/1538-4357/adfccc`. arXiv: `2509.01740 [astro-ph.GA]`.
6. Daniela Calzetti et al. "Quantification of the Age Dependence of Mid-infrared Star Formation Rate Indicators". In: *The Astrophysical Journal* 991.2, 198 (Oct. 2025), p. 198. DOI: `10.3847/1538-4357/adfbe0`. arXiv: `2508.08451 [astro-ph.GA]`.

7. Matteo Correnti et al. "FEAST: Probing the Stellar Population of the Starburst Dwarf Galaxy NGC 4449 with JWST/NIRCam". In: *The Astrophysical Journal* 990.1, 72 (Sept. 2025), p. 72. DOI: 10.3847/1538-4357/adec74. arXiv: 2507.03420 [astro-ph.GA].
8. Alice Knutas et al. "FEAST: JWST Uncovers the Emerging Timescales of Young Star Clusters in M83". In: *The Astrophysical Journal* 993.1, 13 (Nov. 2025), p. 13. DOI: 10.3847/1538-4357/ae018c. arXiv: 2505.08874 [astro-ph.GA].
9. Alex Pedrini et al. "The Near Infrared Spectral Energy Distribution of Young Star Clusters in the FEAST Galaxies: Missing Ingredients at 1–5 μm". In: *The Astrophysical Journal* 992.1, 96 (Oct. 2025), p. 96. DOI: 10.3847/1538-4357/ae0182. arXiv: 2509.01670 [astro-ph.GA].
10. Matthew A. Taylor et al. "A Supermassive Black Hole in a Diminutive Ultracompact Dwarf Galaxy Discovered with JWST/NIRSpec+IFU". In: *Astrophysical Journal Letters* 991.1, L24 (Sept. 2025), p. L24. DOI: 10.3847/2041-8213/ae028e. arXiv: 2503.00113 [astro-ph.GA].

POSTERS

1. Jeremy Sun and **Drew Lapeer**. "Sizes of Embedded Star Clusters from JWST - A Tool for Studying Star Formation". In: *American Astronomical Society Meeting Abstracts*. Vol. 58. American Astronomical Society Meeting Abstracts. Feb. 2026, 443.113, p. 443.113.
2. **Andrew Lapeer** et al. "Do Broad Background Potentials Impact Dynamical SMBH Detection?" In: University of Michigan Undergraduate Symposium. May 2024.
3. **Andrew Lapeer** et al. "Probing the Lower Limits of Detectable Central Black Hole Masses in Virgo Cluster CSS with JWST NIRSpec IFU Kinematics". In: *American Astronomical Society Meeting Abstracts #243*. Vol. 243. American Astronomical Society Meeting Abstracts. Feb. 2024, 110.06, p. 110.06.
4. **Andrew Lapeer** et al. "How Low Can You Go? Lower Limits of Recoverable SMBH Mass in Virgo Cluster UCDs from JWST NIRSpec IFU Data". In: Great Lakes Clusters and Streams Conference. Aug. 2023.
5. **Andrew Lapeer** et al. "How Low Can You Go? Lower Limits of Recoverable SMBH Mass in Virgo Cluster UCDs from JWST NIRSpec IFU Data". In: University of Michigan Undergraduate Symposium. May 2023.

INVITED (†) AND CONTRIBUTED TALKS

2026	† Leiden Observatory Shortlisted Candidate Visit
	AAS 247, Phoenix, Arizona
2025	† University of Wisconsin–Madison Monday Science Seminar
	† Tufts University Astronomy Seminar
	† University of Kansas Lunch Seminar (Virtual),
2023	University of Michigan Undergraduate Research Talks

SELECTED OUTREACH, TEACHING

2025 – · · · ·	**Astronomy on Tap, Western Massachusetts**: Organizing committee
	Astronomy is for Everyone (Public Observing): Volunteer
Summer 2025, 2026	**Research Experience in Astronomy for K-12 Teachers**: Graduate Instructor
2024 – · · · ·	**Astrobites**: Author, Education Co-Chair (2025-), Website Chair (2025-)
Summer 2023, 2024	**Michigan Math and Science Scholars Program**: Undergraduate Instructor
2023 – 2024	**Astronomy Monthly at Washtenaw Community College**: Founder
	University of Michigan Astronomy Department: Undergraduate Lab Instructor
	Student Astronomical Society: Outreach telescope operator

RESEARCH MENTORED STUDENTS

2026 – · · · ·	**Bella Kalina**, University of Massachusetts Undergraduate
	Ada Ciamarra, University of Massachusetts Undergraduate
2024 – 2026	**Jeremy Sun**, University of Massachusetts Undergraduate
2023 – 2024	**Callum Bloom**, University of Michigan Undergraduate

SCIENCE COLLABORATIONS & PROFESSIONAL MEMBERSHIPS

2025 – · · · ·	**Astrobites** (astrobites.org)
2024 – · · · ·	**FEAST-JWST Science Collaboration** (feast-survey.github.io)
2023 – · · · ·	**American Astronomical Society**

SKILLS

Languages	English (Native), Spanish (Conversational/Basic).
Programming	Python, C++, Html, css, LaTeX, Julia
Python Packages	Numpy, Astropy, Matplotlib, SciPy, Photutils, AGAMA, astroML, sklearn, Lmfit, emcee, pPXF.
Tools	DS9, TOPCAT, GALFIT, MAST, SExtractor, FORSTAND, APT, JWST ETC, JWST Pipeline
Observatories	JWST (NIRCam, MIRI, NIRSpec/IFU), HST (WFC3), ALMA (Images, Cubes)